

ADEBANJI OLUWATIMILEYIN ADELOWO

Machine Learning Engineer | Data Scientist | Scientific Computing

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PROFILE

Machine Learning Engineer with a background in mathematical engineering and experience in scientific computing, inverse problems, numerical PDEs, signal processing, and applied machine learning. Combines research experience in optimization and fluid dynamics with industry experience developing ML and LLM systems for aviation telemetry, predictive analytics, and computer vision. Interested in scientific machine learning, computational mathematics, and AI-driven engineering systems.

PROFESSIONAL EXPERIENCE

Data Scientist | Intuos Srl, L'Aquila, Italy

Dec 2024 – Sep 2025

- Developed **ML Classifiers** (Random Forest, LGBMClassifier) for real-time flight phase identification
- Developed **FFT-based signal processing pipelines** for aircraft engine RPM estimation and anomaly detection
- Built REST APIs and automated telemetry data processing pipelines
- Developed **dashboards** for operational monitoring and flight performance analysis

Machine Learning Engineer | N-and Italia Srl, Pognano, Italy

Apr 2023 – Nov 2024

- Developed predictive ML models for vending machine delivery failure forecasting
- Processed and analyzed large-scale operational datasets containing transaction and telemetry logs
- Built dashboards tracking KPIs, user behavior metrics, and machine reliability statistics
- Performed statistical analysis of system performance and operational anomalies

Operational Data Analyst | eResult Srl, Cesena, Italy

Jan – Mar 2023

- Optimized SQL queries and reporting workflows for business intelligence systems
- Improved database performance and automated reporting pipelines

Digital Engineering Intern | Bridgestone EMIA, Rome, Italy

Apr – Oct 2022

- Improved deepcopy efficiency on GeometryMap half-edge data structures in tire cross-section FEA meshing pipelines
- Analysed multi-stage tire assembly build process to identify copy bottlenecks across nested Bézier-backed geometry objects
- Developed geometric algorithms for cubic Bézier curve closest-point computation using scalar minimization

Machine Learning Intern | Fellowship.AI, San Francisco (Remote)

May – Aug 2022

- Researched convolutional networks and Generative Adversarial Networks (GANs)
- Implemented **InsetGAN architecture** for whole-body human image generation; deployed inference service to Swift iOS app via FastAPI

TECHNICAL SKILLS

Programming: Python, SQL, MATLAB, C++, R, $\LaTeX$

ML & AI: PyTorch, TensorFlow, scikit-learn, LangChain, OpenAI API

Data & Engineering: FastAPI, Docker, PostgreSQL, IBM DB2, SQL Server, Git

Signal Processing: FFT analysis, anomaly detection, audio DSP

Numerical Methods: Finite differences, finite elements, finite volumes

Visualization & Cloud: Matplotlib, Seaborn, Plotly, Power BI, React/Vite, AWS, Google Cloud

SELECTED PROJECTS

INTUOS Flight Data Monitoring Dashboard | Full-stack aviation analytics platform; FastAPI + IBM DB2 backend, React/Vite frontend. Real-time telemetry, flight phase classification, engine metrics. *Stack: FastAPI, IBM DB2, React, Vite, Docker*

Aircraft Engine RPM Estimation via Audio DSP | Raspberry Pi system using FFT on microphone audio for RPM extraction; transmitted over UART to ESP32. Validated on Tecnam P92 / Rotax 912.

LLM Engineering Portfolio | Medical RAG Assistant, Enterprise NL-to-SQL, Financial Sentiment Distillation. *Stack: LangChain, OpenAI, Pinecone, FastAPI*

Vending Machine Failure Prediction | Predictive ML for delivery failure forecasting from telemetry logs across 150+ machines.

Image Denoising | ResNet with Residual Channel Attention Blocks and Squeeze-and-Excitation for grayscale denoising; blind denoising with random σ , SSIM+L1 loss, achieving ~ 26 – 28 dB PSNR on LFW.

Multi-Label Image Classification | Pascal VOC 2012 (20 classes) using EfficientNet-B3 with mixed-precision training, OneCycleLR, and mAP evaluation.

Subsurface Skin Scattering | Dipole BSSRDF renderer for all 6 Fitzpatrick skin types in pure PyTorch; analytically derived and measured R_{hemi} across skin tones. *Stack: PyTorch, NumPy*

NeRF vs TensorRF | Pure-PyTorch NeRF and TensorRF trained on a synthetic scene; TensorRF achieves 17.2 dB PSNR / 0.834 SSIM vs NeRF's 14.2 dB / 0.770. *Stack: PyTorch, MPS*

Differentiable PBR Rendering | Cook–Torrance differentiable renderer optimising albedo, surface normals, roughness, and metallic maps from multi-view observations via photometric loss. *Stack: PyTorch, NumPy*

3D Liver Segmentation — Abdominal CT | 3D U-Net trained end-to-end on MSD Task03 liver CT volumes (131 cases). Foreground-biased patch sampling, combined soft-Dice + BCE loss, cosine LR annealing, Gaussian sliding-window inference. Best val Dice **0.9886**, HD95 ≤ 1 mm after 200 epochs on Kaggle T4. *Stack: PyTorch, MONAI, NumPy*

EDUCATION

M.Sc. Mathematical Engineering | University of L'Aquila, Italy | **University of L'Aquila International Scholar** 2018 – 2021

B.Sc. Mathematics | Obafemi Awolowo University, Nigeria 2014 – 2018

HONORS AND AWARDS

University of L'Aquila International Scholarship | University of L'Aquila 2018 – 2021

Professor S.S. Okoya Prize | Excellence in Applied Mathematics, OAU 2018

MTN Foundation Scholarship | Merit-based undergraduate scholarship 2014 – 2018

PTDF Scholarship | Petroleum Technology Development Fund, Nigeria 2014 – 2018

Murli T. Chellaram Foundation Scholarship | Academic excellence award 2014 – 2018